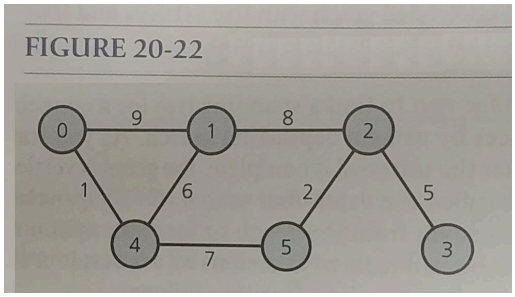
Whit Allee
4/30/2026
COSC-2436-002
Prof. Alexander Katrompas

Asg-9 Graph Theory

1. Give the adjacency matrix and adjacency list for:

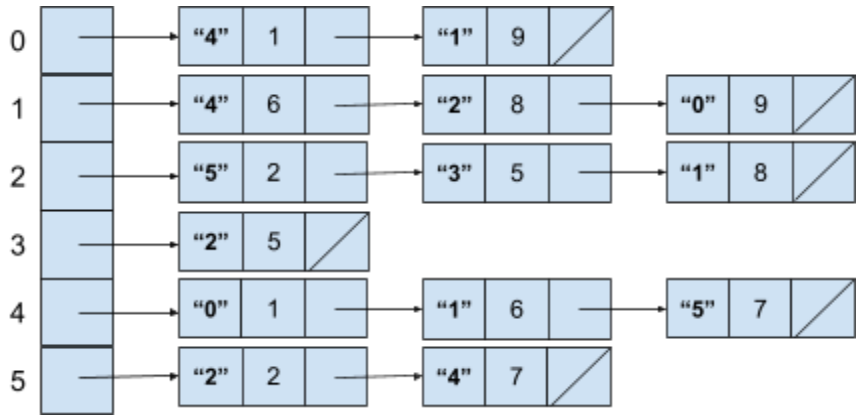
a. Figure 20-22



Matrix:

	0	**1**	**2**	**3**	**4**	**5**
0	∞	9	∞	∞	1	∞
1	9	∞	8	∞	6	∞
2	∞	8	∞	5	∞	2
3	∞	∞	5	∞	∞	∞
4	1	6	∞	∞	∞	∞
5	∞	∞	2	∞	∞	∞

List:



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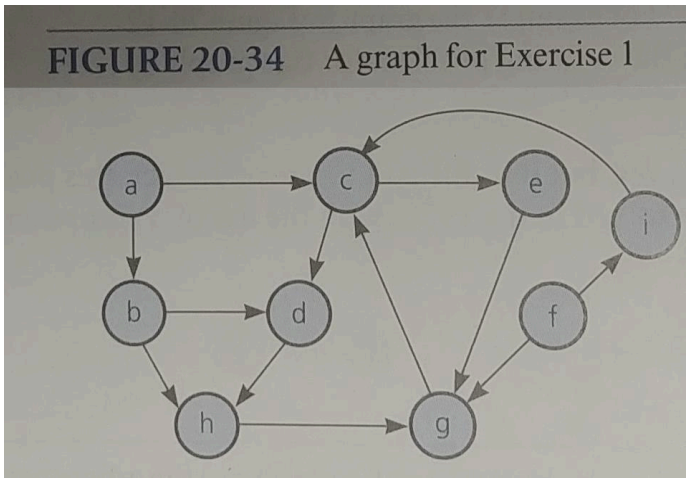
b. Figure 20-34



Matrix:

	a	b	c	d	e	f	g	h	i
a	0	1	1	0	0	0	0	0	0
b	0	0	0	1	0	0	0	1	0
c	0	0	0	1	1	0	0	0	0
d	0	0	0	0	0	0	0	1	0
e	0	0	0	0	0	0	1	0	0
f	0	0	0	0	0	0	1	0	1
g	0	0	1	0	0	0	0	0	0
h	0	0	0	0	0	0	1	0	0
i	0	0	1	0	0	0	0	0	0

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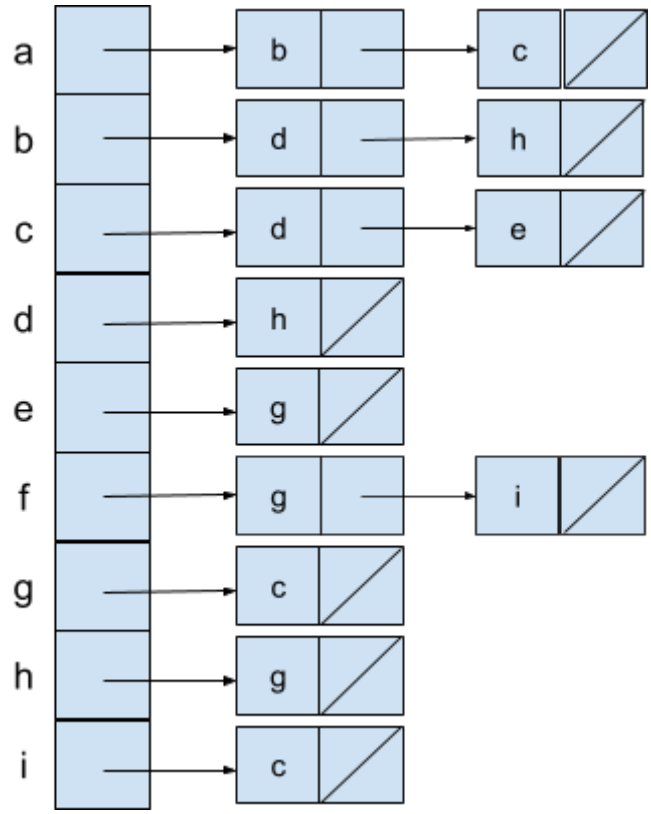
List:



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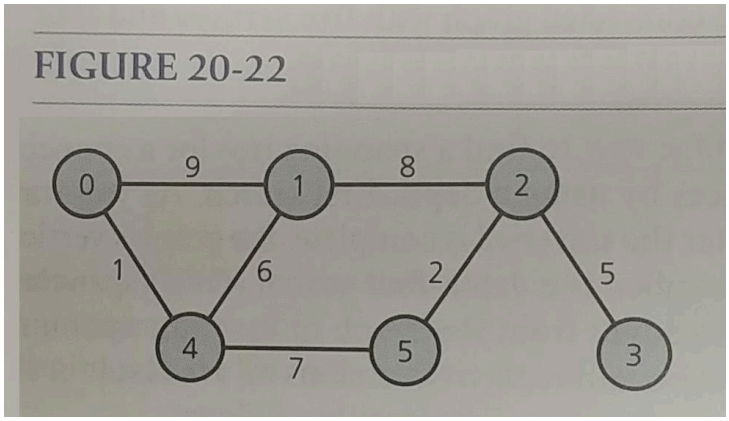
2. Use both the depth-first and breadth-first strategies to traverse the graph in Figure 20-22 beginning with vertex 0. Do the same for Figure 20-36, beginning with vertex a. List the vertices in the order in which each traversal visits them.

Figure 20-22



DFS:

0 -> 1 -> 2 -> **3**
back -> 2 -> 5 -> **4**
back -> **5**
back -> **2**
back -> **1**
back -> **0**

Answer: 3, 4, 5, 2, 1, 0

BFS:

0 -> 1, 4
1 -> 2
4 -> 5
2 -> 3
5
3

Answer: 0, 1, 4, 2, 5, 3

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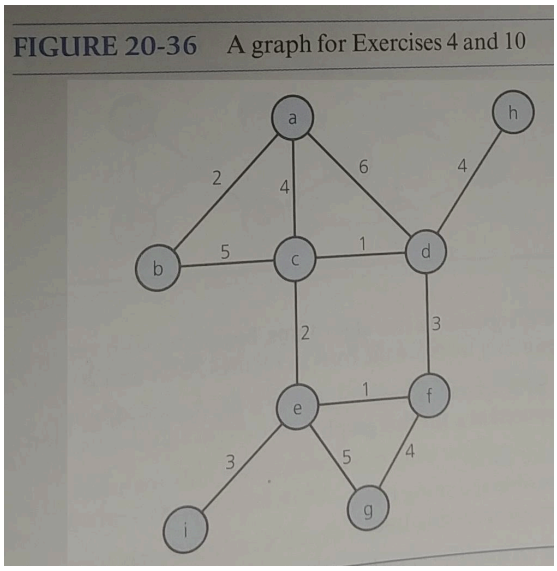
Figure 20-36



DFS:

a -> b -> c -> d -> f -> e -> **g**
back -> e -> **i**
back -> **e**
back -> **f**
back -> d -> **h**
back -> **d**
back -> **c**
back -> **b**
back -> **a**

Answer: g, i, e, f, h, d, c, b, a

BFS:

a -> b, c, d
b
c -> e
d -> f, h
e -> g, i
f
h
g
i

Answer: a, b, c, d, e, f, h, g, i

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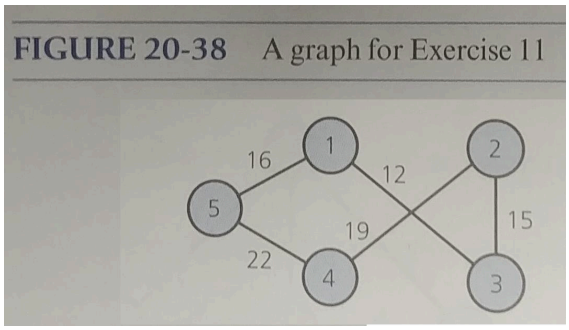
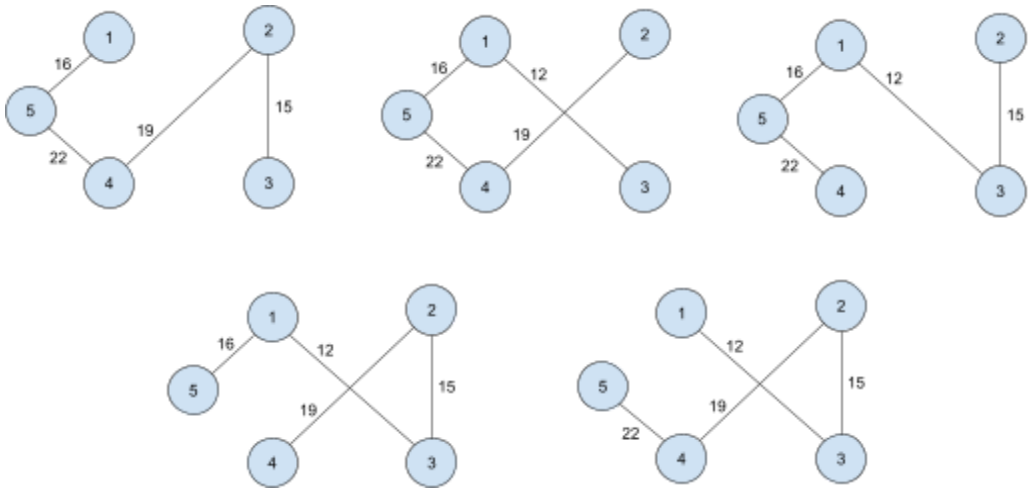
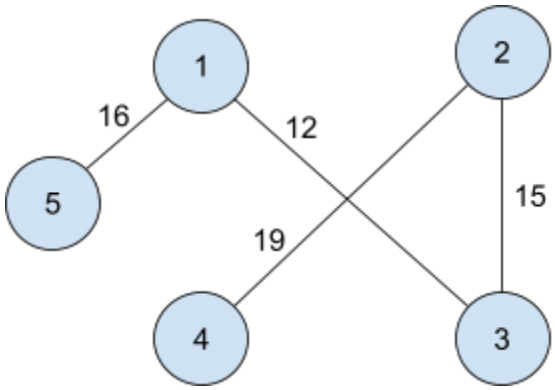
3. For the graph 20-38:



a. Draw all possible spanning trees.



b. Draw the minimum spanning tree.



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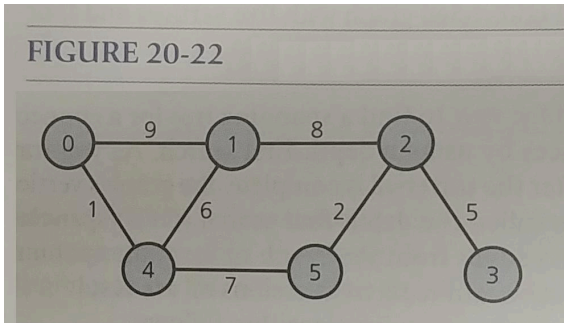
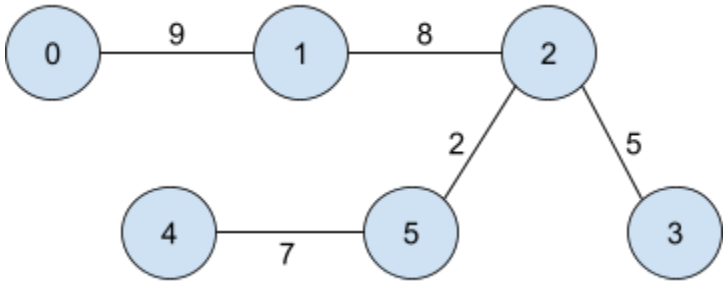
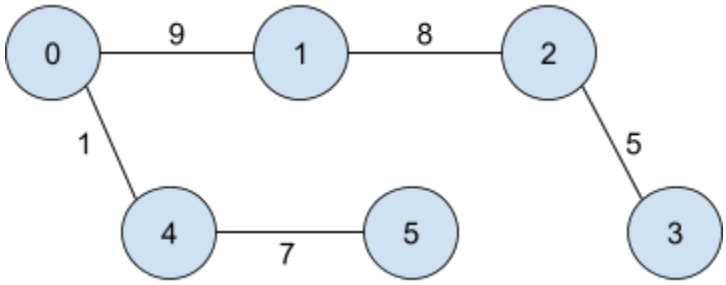
4. For Figure 20-22:



a. Draw the DFS spanning tree whose root is vertex 0.



b. Draw the BFS spanning tree whose root is vertex 0.